

SQL Cheat Sheet

1. Database Basics

```
CREATE DATABASE School; USE School; DROP DATABASE School;
```

2. Tables

```
CREATE TABLE Students ( StudentID INT PRIMARY KEY, Name VARCHAR(100), Age INT, Grade VARCHAR(10) ); DROP TABLE Students; ALTER TABLE Students ADD Email VARCHAR(100); ALTER TABLE Students DROP COLUMN Age;
```

3. Insert Data

```
INSERT INTO Students (StudentID, Name, Age, Grade) VALUES (1, 'Ali Ahmed', 20, 'A'); INSERT INTO Students (StudentID, Name, Age, Grade) VALUES (2, 'Sara Mohamed', 22, 'B'), (3, 'Omar Hassan', 21, 'A');
```

4. Select Data

```
SELECT * FROM Students; SELECT Name, Grade FROM Students; SELECT * FROM Students WHERE Grade = 'A'; SELECT * FROM Students WHERE Age > 20; SELECT * FROM Students WHERE Grade = 'A' AND Age > 20;
```

5. Sorting & Limiting

```
SELECT * FROM Students ORDER BY Age ASC; SELECT * FROM Students LIMIT 2;
```

6. Update & Delete

```
UPDATE Students SET Grade = 'B' WHERE StudentID = 1; DELETE FROM Students WHERE StudentID = 3;
```

7. Aggregate Functions

```
SELECT COUNT(*) FROM Students; SELECT AVG(Age) FROM Students; SELECT MAX(Age) FROM Students; SELECT MIN(Age) FROM Students; SELECT SUM(Age) FROM Students;
```

8. GROUP BY & HAVING

```
SELECT Grade, COUNT(*) AS StudentCount FROM Students GROUP BY Grade; SELECT Grade, COUNT(*) AS StudentCount FROM Students GROUP BY Grade HAVING COUNT(*) > 1;
```

9. Joins

```
CREATE TABLE Courses ( CourseID INT PRIMARY KEY, CourseName VARCHAR(100),  
StudentID INT ); SELECT s.Name, c.CourseName FROM Students s INNER JOIN Courses c  
ON s.StudentID = c.StudentID; SELECT s.Name, c.CourseName FROM Students s LEFT  
JOIN Courses c ON s.StudentID = c.StudentID;
```

10. Subqueries

```
SELECT * FROM Students WHERE Age > (SELECT AVG(Age) FROM Students);
```

11. Constraints

```
CREATE TABLE Teachers ( TeacherID INT PRIMARY KEY, Name VARCHAR(100) NOT NULL,  
Email VARCHAR(100) UNIQUE );
```

12. Indexes

```
CREATE INDEX idx_name ON Students(Name);
```

13. Views

```
CREATE VIEW A_Students AS SELECT Name, Grade FROM Students WHERE Grade = 'A';  
SELECT * FROM A_Students;
```

14. Transactions (TCL)

```
START TRANSACTION; UPDATE Students SET Grade = 'C' WHERE StudentID = 2; ROLLBACK;  
COMMIT;
```

15. User & Permissions (DCL)

```
CREATE USER 'ahmed'@'localhost' IDENTIFIED BY '1234'; GRANT SELECT, INSERT ON  
School.* TO 'ahmed'@'localhost'; REVOKE INSERT ON School.* FROM  
'ahmed'@'localhost';
```